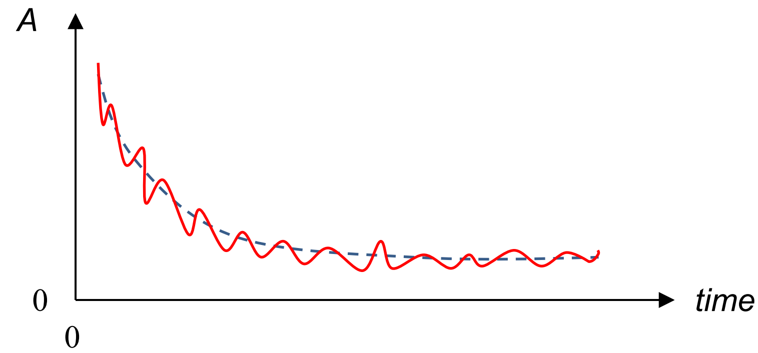
5 **(a)** Must draw a graph (A or N against *time*) that shows jagged lines and a vertical displacement.



Random decay – causes the jagged lines over the smooth best fit line

Background radiation – causes the graph to be displaced upwards by a constant value

(b) Assuming N remains constant over the 50s duration

$$A = \frac{880+885+879+878+886}{5(10)}$$

$$= 88.16 \text{ Bq}$$

$$A = \lambda N$$

$$88.61 = \frac{\ln 2}{14.6(60)(60)(24)} N$$

$$N = 1.60 \times 10^{8}$$

(c) At t=0,
$$\frac{A_{32}}{A_{33}} = \frac{0.80}{0.20} \quad \cdots\cdots(1)$$

After t days,
$$\frac{A_{32}'}{A_{33}'} = \frac{0.30}{0.70} \quad \cdots\cdots(2)$$

$$\frac{A_{32}'}{A_{33}'} = \frac{A_{32}\, e^{-\lambda_1 t}}{A_{33}\, e^{-\lambda_2 t}}$$

$$\frac{0.30}{0.70} = \frac{0.80\, e^{-\lambda_1 t}}{0.20\, e^{-\lambda_2 t}}$$

$$\left(\frac{3}{7}\right)\left(\frac{2}{8}\right) = e^{(\lambda_2 - \lambda_1)t}$$

$$0.107 = e^{\left(\frac{\ln 2}{25.8} - \frac{\ln 2}{14.6}\right)t} \qquad \textit{where} \quad \lambda_1 = \frac{\ln 2}{14.6} \quad \lambda_2 = \frac{\ln 2}{25.8}$$

$$0.107 = e^{-0.02061\,t}$$

$$t = 108 \text{ days}$$